

Vignesh Thanasekaran

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ABOUT

Driven by a strong interest in embedded systems, I am passionate about working at the intersection of hardware and software. I enjoy exploring low-level programming, real-time systems, and building efficient, reliable solutions that solve real-world problems. Currently developing firmware for precision metering instruments, I bring hands-on experience in STM32 microcontrollers, serial communication protocols, and embedded system design. Continuously learning and staying curious are at the core of my approach to engineering.

EXPERIENCE

Junior Embedded Firmware Engineer

Signal and Systems Pvt. Ltd.

May 2025 – Present

Chennai, Tamil Nadu

- Developed Smart Calibration firmware for the Three-Phase Portable Reference Standard Meter (TPRS), automating high-accuracy calibration workflows on STM32 microcontrollers.
- Developed firmware for TPRS and SPRS (Single-Phase Portable Reference Standard Meter) platforms used in precision energy meter testing and certification.
- Implemented and debugged peripheral communication on STM32 using SPI, I2C, and UART protocols for sensor interfacing, data acquisition, and display control.
- Authored user manuals and technical documentation using Sphinx, improving field deployment and end-user support quality.
- Collaborated with hardware and QA teams to perform integration testing and validate firmware against IEC metrological accuracy standards.

TECHNICAL SKILLS

Programming Languages: C, C++, Embedded C, Python

Microcontrollers & Architecture: STM32 (ARM Cortex-M), LPC2129

Communication Protocols: UART, SPI, I2C, CAN, TCP

Tools & IDEs: STM32CubeIDE, Keil µVision, Flash Magic, Sphinx

Operating Systems: Windows, Linux

Domain: Embedded Systems, Firmware Development, Precision Metering, Technical Documentation

EDUCATION

Saranathan College of Engineering

Bachelor of Engineering in Electronics and Communication | Cum. GPA: 8.67/10.0

Trichy, Tamil Nadu

Aug 2020 – May 2024

S F S Matriculation Higher Secondary School

Higher Secondary (Class XII) | Percentage: 73.83%

Pudukkottai, Tamil Nadu

Jun 2018 – Apr 2020

S F S Matriculation Higher Secondary School

SSLC (Class X) | Percentage: 89.8%

Pudukkottai, Tamil Nadu

Jun 2017 – Apr 2018

PROJECTS

Smart Calibration – TPRS Calibration Process

Embedded Firmware Project

STM32, Embedded C, SPI, UART, ADC

- Independently developed Smart Calibration firmware to automate the calibration sequence of the Three-Phase Portable Reference Standard Meter (TPRS), reducing manual calibration effort.
- Implemented high-resolution ADC sampling and calibration algorithms on STM32 with precise data exchange over SPI and UART interfaces.
- Performed integration testing to verify calibration accuracy and compliance with IEC metrological standards.

TPRS & SPRS – Portable Reference Standard Meters

Embedded Firmware Project

STM32, Embedded C, SPI, UART, Sphinx

- Developed firmware modules for Three-Phase (TPRS) and Single-Phase (SPRS) Portable Reference Standard Meters for precision energy meter calibration.
- Integrated multiple peripherals using SPI, I2C, and UART protocols for data acquisition, display management, and system control.
- Produced and maintained user manuals and technical documentation using Sphinx documentation generator.

Vehicle Control Dashboard using CAN Protocol

Dec 2024 – Jan 2025

Embedded System Project

LPC2129, Embedded C, CAN Bus

- Designed and implemented a multi-node vehicle dashboard for real-time monitoring of fuel level, turn indicators, and wiper control using CAN protocol.
- Programmed four CAN nodes (Fuel Control, Indicator, Wiper, Main Node) on LPC2129 microcontrollers with LCD display output and real-time data synchronisation.

IoT-Based VR Home Automation with LPG Gas Management

Jan 2024 – May 2024

IoT Based Project

Python, IoT, Virtual Reality, CNN

- Developed a VR-based home automation interface integrated with IoT sensors for real-time device control and LPG gas level monitoring.
- Implemented CNN-based hand gesture recognition for intuitive, touch-free control within the virtual environment.

CERTIFICATIONS

- Embedded Systems | Vector India (*Completed: 2025*)
- C Programming | HackerRank (*Certified*)
- IoT Professional Course | Naan Mudhalvan

ACHIEVEMENTS

- Graduated with First Class Distinction – GPA 8.67 / 10.0
- Participated in Smart India Hackathon (SIH) 2022
- Runner-up in Inter-College Cricket & Hockey Competitions – Saranathan College of Engineering, Trichy (2024)

SOFT SKILLS

- Communication
- Problem-Solving
- Time Management
- Adaptability
- Attention to Detail

LANGUAGES

- English – Professional Working Proficiency
- Tamil – Native